TITLE

May 9, 2026

Contents

1. Preliminaries on schemes

1.1. Dedekind schemes and fibers.

Definition 1.1.1 (Dedekind scheme)**.** A *Dedekind scheme* is a scheme S that is integral (Definition 1.2.5), Noetherian, normal, and of Krull dimension (Definition 1.2.8) 1. Equivalently, every point $s \in S$ has an open neighborhood $U = \operatorname{Spec} R$ such that R is a Dedekind domain. In particular, for every point $s \in S$, the local ring $\mathcal{O}_{S,s}$ is either a field (if s is the generic point) or a discrete valuation ring (if s is a closed point).

Proposition 1.1.2. Every DVR is a Dedekind domain. Conversely, every local Dedekind domain is a DVR.

Proof. By definition, every DVR is a PID. All PID's are Noetherian. All PID's are also UFD's which are in turn integrally closed (Definition A.2.2) in their fields of fractions.

The ideals of a DVR $(R, \mathfrak{m})$ are of the form $\mathfrak{m}^n$ for some integer $n \geq 0$. The ideal $\mathfrak{m}^0$ is the unit ideal. The ideal $\mathfrak{m}^1$ is the maximal ideal. For $n \geq 2$, the ideal $\mathfrak{m}^n$ is not a prime ideal — fixing a uniformizer π of R, the element $\pi^n = \pi^{n-1} \cdot \pi$ is an element of $\mathfrak{m}^n$ and yet neither π^{n-1} nor π is an element of $\mathfrak{m}^n$.

We have thus shown that R is Noetherian, integrally closed closed whose only prime ideal is the maximal ideal, i.e. R is a Dedekind domain.

Conversely, suppose that R is a local Dedekind domain. Let $\mathfrak{m}$ be its maximal ideal.

(♠ TODO: verify the following converse argument) Since R is Noetherian and every nonzero prime ideal is maximal, R has Krull dimension at most 1. Because R is local, $\mathfrak{m}$ is the unique nonzero prime ideal. We first show that $\mathfrak{m}$ is principal. Let K be the field of fractions of R, and consider the fractional ideal

$$I = \{y \in K \mid y\mathfrak{m} \subseteq R\}.$$

Clearly $R \subseteq I$. Since $\mathfrak{m}$ contains a nonzero element and R is a domain, $I \neq (0)$. Moreover, for any $y \in I$, we have $y\mathfrak{m} \subseteq R$, which implies $y^2\mathfrak{m} = y(y\mathfrak{m}) \subseteq yR \subseteq I$ by induction;

thus I is actually a subring of K. Because $\mathfrak{m}$ is finitely generated (as R is Noetherian), say $\mathfrak{m} = (a_1, \dots, a_n)$, and because $ya_i \in R$ for all i, we have $xI \subseteq R$ for any nonzero $x \in \mathfrak{m}$ (since $xy = yx \in y\mathfrak{m} \subseteq R$). Hence $I \subseteq x^{-1}R$. As R is Noetherian, the submodule I of the free module $x^{-1}R \cong R$ is finitely generated. For any $y \in I$, multiplication by y maps the finitely generated faithful R-module I into itself, so by the determinant trick (or Cayley-Hamilton), y satisfies a monic polynomial over R. Thus every element of I is integral over R. Since R is integrally closed in K, we conclude $I = R$. This means $\mathfrak{m}^{-1}\mathfrak{m} = R$, so $\mathfrak{m}$ is an invertible ideal.

In a local ring, every invertible ideal is principal (this follows immediately from Nakayama's Lemma: if $\mathfrak{m}/\mathfrak{m}^2$ required more than one generator over $R/\mathfrak{m}$, invertibility would fail). Hence $\mathfrak{m} = (\pi)$ for some uniformizer π.

Now let $J \neq (0)$ be any nonzero ideal of R. Since $\dim R \leq 1$, the only prime ideals are (0) and $\mathfrak{m}$, so $\sqrt{(x)} = \mathfrak{m}$ for every $0 \neq x \in J$. Thus there exists a largest integer $n \geq 0$ such that $J \subseteq \mathfrak{m}^n = (\pi^n)$. Writing $J = \pi^n K$ for some ideal K, we have $K \not\subseteq \mathfrak{m}$, which forces $K = R$ (as $\mathfrak{m}$ is the unique maximal ideal). Therefore $J = (\pi^n)$, proving that every nonzero ideal of R is principal. Consequently, R is a local PID. A local PID with Krull dimension 1 is precisely a discrete valuation ring by definition. This completes the proof.

□

Definition 1.1.3 (Special fiber and generic fiber)**.** Let S be a integral (Definition 1.2.5) scheme, let η be its generic point (Definition 1.1.4), and let $K = \mathcal{O}_{S,\eta}$ be the function field of S. Let $s \in S$ be a closed point.

1. For any morphism of schemes (Definition 1.2.1) $f : X \to S$, the *generic fiber of X over S* (or simply the *generic fiber*) is the base change (Definition A.2.5):
$$X_\eta = X \times_S \operatorname{Spec} K$$
of f along $\operatorname{Spec} K \to S$.
2. For any morphism of schemes $f : X \to S$, the *special fiber of X at s* is the base change (Definition A.2.5):
$$X_s = X \times_S \operatorname{Spec} k(s)$$
of f along $\operatorname{Spec} k(s) \to S$, where $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$ is the residue field (Definition 1.2.9) at s.

   In the case that S has only closed point (e.g. S is the ring of integers of a local field), the *special fiber of X* refers to the special fiber of X at this unique closed point.

Definition 1.1.4. Let X be a topological space. A point $\eta \in X$ is called a *generic point of X* if the closure of the singleton set $\{\eta\}$ is the entire space X, i.e., $\overline{\{\eta\}} = X$. More generally, if Z is an irreducible closed subset of X, a point $\eta \in Z$ is called a *generic point of Z* if $\overline{\{\eta\}} = Z$. In the context of schemes, every irreducible closed subset has a unique generic point.

Definition 1.1.5 (Closed point of a topological space)**.** Let X be a topological space. A point $x \in X$ is called a *closed point of X* if the singleton set $\{x\}$ is closed in the topology of X, i.e., its closure satisfies $\overline{\{x\}} = \{x\}$.

In a T_1 space, every singleton set is closed, so every point is a closed point.

1.2. **Basic scheme-theoretic notions.**

Definition 1.2.1 (Morphism of schemes)**.** Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be schemes. A *morphism of schemes* is a morphism as locally ringed spaces.

In particular, there is a category, often denoted by Sch, $\mathbf{Sch}$ etc., whose objects are schemes and whose morphisms are morphisms of schemes.

Definition 1.2.2. Let $f : X \to Y$ be a morphism of schemes (Definition 1.2.1). The morphism f is said to be *proper* if it satisfies the following three conditions:

1. f is separated,
2. f is of finite type,
3. f is universally closed.

Definition 1.2.3 (Smooth Morphism of Schemes)**.** Let $f : X \to S$ be a morphism (Definition 1.2.1) of schemes.

We say that f is *smooth*, and that X is a *smooth scheme over S*, if it satisfies the following conditions:

- f is locally of finite presentation: for every point $x \in X$, there exists an open neighborhood $U \subseteq X$ of x and an open neighborhood $V \subseteq S$ of $f(x)$ such that the restriction $f|_U : U \to V$ corresponds to a morphism of rings $\mathcal{O}_S(V) \to \mathcal{O}_X(U)$ that is finitely presented.
- f is flat: the induced map on local rings is flat.
- For every point $x \in X$, the fiber $X_{f(x)} = X \times_S \operatorname{Spec} \kappa(f(x))$ is a smooth variety over the residue field (Definition 1.2.9) $\kappa(f(x))$, equivalently, the sheaf of relative Kähler differentials $\Omega_{X/S}$ is locally free of finite rank.

Informally, a smooth morphism behaves like a submersion in differential geometry, providing "nice" fiber structures and descent properties.

Given a scheme S, the *category of smooth schemes over S* is the following locally small category:

- The objects are smooth morphisms $X \to S$.
- The morphisms betewen objects $X_1 \to S$ and $X_2 \to S$ are S-morphisms $X_1 \to X_2$ such that the following commutes:

$$\begin{tikzcd} X_1 \arrow[rr] \arrow[dr] & & X_2 \arrow[dl] \\ & S & \end{tikzcd}$$

The category of smooth schemes over S is often denoted by notations such as Sm/S, $\mathbf{Sm}/S$, Sm_S, $\mathbf{Sm}_S$ etc.

Definition 1.2.4 (Affine scheme). Let A be a commutative ring with unity. Define the set $\mathrm{Spec}(A)$ to be the set of all prime ideals of A. Equip it with the *Zariski topology*, which is the topology whose closed sets are given by *vanishing loci*

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec}(A) : I \subseteq \mathfrak{p}\}$$

for ideals $I \subseteq A$. Define the sheaf $\mathcal{O}_{\mathrm{Spec}(A)}$, called the *structure sheaf of* Spec A, by

$$\mathcal{O}_{\mathrm{Spec}(A)}(U) = \{ \text{ locally defined fractions of elements of } A \text{ on } U \ \},$$

for each open set $U \subseteq \mathrm{Spec}(A)$. More precisely, $\mathcal{O}_{\mathrm{Spec}(A)}(U)$ can be described as the set of functions $s : U \to \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}}$ such that

1. For each $\mathfrak{p} \in U$, the value $s(\mathfrak{p})$ lies in the localization $A_{\mathfrak{p}}$
2. For each $\mathfrak{p} \in U$, there exists an open neighborhood $V \subseteq U$ containing $\mathfrak{p}$ and elements $a, f \in A$ such that for all $\mathfrak{q} \in V$,
   - $f \notin \mathfrak{q}$, so that the fraction $\frac{a}{f}$ is well-defined in $A_{\mathfrak{q}}$.
   - $s(\mathfrak{q}) = \frac{a}{f}$ in $A_{\mathfrak{q}}$.

It is the case that the stalk at $\mathfrak{p} \in \mathrm{Spec}(A)$ is canonically the localization $A_{\mathfrak{p}}$. Then $(\mathrm{Spec}(A), \mathcal{O}_{\mathrm{Spec}(A)})$ is a locally ringed space, called the *affine scheme associated to A*.

Moreover, given $f \in A$, we define the locus $D(f)$ by

$$D(f) = \mathrm{Spec}\, A \setminus V((f)) = \{\mathfrak{p} \in \mathrm{Spec}\, A : f \notin \mathfrak{p}\}$$

Definition 1.2.5. Let X be a scheme. We say that X is an *integral scheme* if X is both a reduced scheme and an irreducible topological space. Equivalently, X is integral if for every non-empty open subset $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is an .

Definition 1.2.6 (Locally Noetherian Scheme and Noetherian Scheme). Let X be a scheme.

- X is called *locally Noetherian* if it admits an open cover $\{U_i\}$ such that for each i, the ring $\mathcal{O}_X(U_i)$ of regular functions on U_i is a Noetherian ring. Equivalently, X is locally Noetherian if it is covered by open affine subschemes $\mathrm{Spec}\, A_i$ with each A_i a Noetherian ring.
- X is called *Noetherian* if it is locally Noetherian and quasi-compact, i.e., X can be covered by finitely many affine opens $\mathrm{Spec}\, A_i$ where each A_i is Noetherian.

Definition 1.2.7. Let X be a scheme.

- We say that X is *normal at a point $x \in X$* if the local ring $\mathcal{O}_{X,x}$ is an integrally closed (Definition A.2.2) domain.
- The scheme X is called *normal* if it is normal at every point $x \in X$. If X is integral (Definition 1.2.5), this is equivalent to saying that for every affine open subset $U = \mathrm{Spec}\, A$, the ring A is an integrally closed domain.
- We say that X is *factorial at a point $x \in X$* if the local ring $\mathcal{O}_{X,x}$ is a UFD.
- The scheme X is called *factorial* if it is factorial at every point.

Definition 1.2.8. Let X be a topological space. The *dimension of X* or *Krull dimension of X*, denoted by $\dim(X)$, is the supremum of the lengths of strictly ascending chains of distinct irreducible closed subsets of X. That is, $\dim(X)$ is the supremum of integers n such that there exists a chain

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n$$

where each Z_i is a non-empty irreducible closed subset of X. If X is empty, its dimension is defined to be $-\infty$ (or sometimes -1).

Definition 1.2.9. Let X be a scheme and let $x \in X$ be a point. The *residue field of the scheme X at the point x* is defined as

$$\kappa(x) = \kappa(\mathcal{O}_{X,x}) := \mathcal{O}_{X,x}/\mathfrak{m}_x.$$

where $\mathcal{O}_{X,x}$ is the local ring of X at x with maximal ideal $\mathfrak{m}_x$.

When X is a scheme over a field (Definition A.1.3) k, the residue field of X at x is often denoted by $k(x)$.

Definition 1.2.10. Let R be an integral domain. Let $n \geq 1$ be an integer. The *function field over R in n variables $x_1, \ldots, x_n$* is the field (Definition A.1.3) $R(x_1, \ldots, x_n)$ defined as the field of fractions of the polynomial ring $R[x_1, \ldots, x_n]$, which is an integral domain .

$R(x_1, \ldots, x_n)$ is also called the *field of rational numbers over R in n variables*. An element of $R(x_1, \ldots, x_n)$ is called a *rational function over R in n variables*.

2. Algebraic group schemes and abelian varieties

2.1. Group schemes.

Definition 2.1.1. Let S be a scheme. An algebraic group scheme over S (or an S-group scheme) is a group object G in the category of schemes over S; that is, G is an S-scheme equipped with S-morphisms: $m : G \times_S G \to G$ (*multiplication*), $i : G \to G$ (*inverse*), and $e : S \to G$ (*identity*/*identity section*), satisfying the group axioms expressed by the commutativity of the following diagrams:

1. **Associativity**
$$\begin{tikzcd} G \times_S G \times_S G \arrow[r, "m\times\mathrm{id}"] \arrow[d, "\mathrm{id}\times m"'] & G \times_S G \arrow[d, "m"] \\ G \times_S G \arrow[r, "m"] & G \end{tikzcd}$$

2. **Identity**
$$\begin{tikzcd} G \times_S S \arrow[r, "\mathrm{id}\times e"] \arrow[dr, "\simeq"'] & G \times_S G \arrow[d, "m"] \\ & G \end{tikzcd} \qquad \begin{tikzcd} S \times_S G \arrow[r, "e\times\mathrm{id}"] \arrow[dr, "\simeq"'] & G \times_S G \arrow[d, "m"] \\ & G \end{tikzcd}$$

3. **Inverse**
$$\begin{tikzcd} G \arrow[r, "(\mathrm{id},i)"] \arrow[d, "\mathrm{id}"'] & G \times_S G \arrow[d, "m"] \\ G \arrow[r, "e\circ\pi"] & G \end{tikzcd}$$
where $\pi : G \to S$ is the structure morphism and $e \circ \pi$ sends g to the identity section.

If the multiplication map also satisfies the commutative axiom, then the group scheme G is called *commutative* or *abelian*.

If G is affine over S, we call it an *affine group scheme over S*.

If the base scheme S is the spectrum of a field k, then we call G a *k-algebraic group* or an *algebraic group (scheme) over k*. If G is additionally a k-variety, then we call G a *k-group variety*.

Definition 2.1.2.
1. Let k be a field (Definition A.1.3), and let G be a group scheme (Definition 2.1.1) over $\operatorname{Spec}(k)$ (Definition 1.2.4). The *identity component or neutral component of G*, denoted G^0, is the unique connected component of G containing the identity point $e \in G(k)$.
2. Let S be a scheme, and let G be an algebraic group scheme over S. The *identity component of G*, denoted G^0, is the unique open and closed subgroup scheme of G such that for every geometric point $\bar{s} \to S$, the fiber $(G^0)_{\bar{s}}$ is the connected component of the identity element in the group scheme $G_{\bar{s}}$.

   Note that G^0/S might not be connected if S is not connected

Proposition 2.1.3 (see [Mil17, Between Definition 6.9 and Proposition 6.10])**.** Let G/k be a smooth algebraic group scheme over a field. The strong identity component G^{so} coincides with the identity component G^0 (Definition 2.1.2).

Definition 2.1.4. (♠ TODO: TODO: define a (commutative) group scheme) (♠ TODO: TODO: define the multiplicative group scheme) Let S be a scheme. The *additive group scheme over S* is the group scheme $\mathbb{G}_a = \mathbb{G}_{a,S}$ over S defined by

$$\mathbb{G}_{a,S} = \operatorname{Spec} \mathcal{O}_S[T]$$

with group structure morphisms:

- **Comultiplication (Addition):** The morphism $\Delta : \mathbb{G}_{a,S} \to \mathbb{G}_{a,S} \times_S \mathbb{G}_{a,S}$ corresponding to the ring homomorphism $\mathcal{O}_S[T] \to \mathcal{O}_S[T] \otimes_{\mathcal{O}_S} \mathcal{O}_S[T]$ given by $T \mapsto T \otimes 1 + 1 \otimes T$.
- **Counit (Identity):** The morphism $\varepsilon : \mathbb{G}_{a,S} \to S$ corresponding to $T \mapsto 0$.
- **Coinverse (Inversion):** The morphism $\iota : \mathbb{G}_{a,S} \to \mathbb{G}_{a,S}$ corresponding to $T \mapsto -T$.

Thus, $(\mathbb{G}_{a,S}, \Delta, \varepsilon, \iota)$ is a commutative group scheme over S. Note that we may speak of the *additive group scheme over a ring R* as the additive group scheme over $\operatorname{Spec} R$.

Definition 2.1.5 (Multiplicative group scheme)**.** Let S be a scheme. The *multiplicative group scheme over S*, denoted $\mathbb{G}_{m,S}$, is the group scheme over S defined as the open subscheme

$$\mathbb{G}_{m,S} := \mathbb{A}^1_S \setminus \{0\}_S$$

of the affine line $\mathbb{A}^1_S$, equipped with the group law given by multiplication of functions:

$$m : \mathbb{G}_{m,S} \times_S \mathbb{G}_{m,S} \to \mathbb{G}_{m,S}, \quad (x, y) \mapsto xy.$$

The identity section is the morphism

$$e : S \to \mathbb{G}_{m,S}, \quad s \mapsto 1,$$

and the inversion morphism is given by

$$i : \mathbb{G}_{m,S} \to \mathbb{G}_{m,S}, \quad x \mapsto x^{-1}.$$

Definition 2.1.6 (Kernel of a homomorphism of group schemes over a scheme)**.** Let S be a scheme and let $\phi\colon G_1 \to G_2$ be a homomorphism of algebraic group schemes over S.

The *kernel of ϕ*, denoted by $\ker\phi$ or $A[\phi]$, is the fiber of ϕ at the identity section (Definition 2.1.1) $e_2\colon S \to G_2$ of G_2. Equivalently, $\ker\phi$ is the subgroup scheme of G_1 defined as the pullback:

$$\ker\phi = G_1 \times_{G_2} S,$$

(♠ TODO: finite group scheme) where the map $G_1 \to G_2$ is ϕ and the map $S \to G_2$ is the identity section. If ϕ is a finite morphism, then $\ker\phi$ is a finite group scheme over S.

Definition 2.1.7. Let S be a scheme, and let G be an algebraic group scheme over S. A *subgroup scheme* (or synonymously an *algebraic subgroup*) H of G is an S-subscheme

$$H \subseteq G$$

such that for every S-scheme T, the subset $H(T) \subseteq G(T)$ is a subgroup of the group $G(T)$. Equivalently, H is a group object in the category of S-schemes equipped with a monomorphism of group schemes $H \to G$.

Definition 2.1.8 (Quotient of an algebraic group scheme by an algebraic subgroup)**.** Let S be a scheme. Let G be an algebraic group scheme over S (Definition 2.1.1), and let $H \subseteq G$ be a closed subgroup scheme over S, i.e., a monomorphism of group schemes $H \to G$ over S.

The *quotient sheaf* G/H is the fppf sheafification of the presheaf

$$T \mapsto G(T)/H(T)$$

on the category of S-schemes, where T is any S-scheme.

If G/H is representable by a scheme (or by an algebraic space) Q over S, then Q is called the *quotient of G by H*. In general, if H is normal, then G/H inherits the structure of a group scheme over S.

Otherwise, G/H is just a sheaf of sets (or algebraic space on $(\mathrm{Sch}/S)_{fppf}$) over S without a natural group structure.

2.2. **Algebraic tori and semi-abelian group schemes.**

Definition 2.2.1 (Algebraic torus)**.** An *(algebraic) torus over a field k* is an algebraic group (Definition 2.1.1) T over k such that the base change $T_{\overline{k}} = T \times_k \overline{k}$ is isomorphic as an algebraic group over $\overline{k}$ to $(\mathbb{G}_m)^n$ (Definition 2.1.5) for some nonnegative integer n.

Equivalently, an algebraic torus over k is an algebraic torus over $\operatorname{Spec} k$ (Definition 1.2.4).

Definition 2.2.2 (Semi-abelian group scheme over a scheme)**.** Let S be a scheme. A *semi-abelian (group) scheme over S* is an algebraic group scheme (Definition 2.1.1) $\pi\colon G \to S$ that fits into an extension of an abelian scheme by an algebraic torus, i.e., there exists a short exact sequence of algebraic groups over S:

$$1 \to T \to G \to A \to 0,$$

where $\pi_T\colon T \to S$ is an algebraic torus over S, $\pi_A\colon A \to S$ is an abelian scheme over S (Definition 4.0.1), and the maps are homomorphisms of algebraic group schemes over S. If T has rank r, we say that G has *torus rank r*. When $r = 0$, the semi-abelian group scheme is simply an abelian scheme.

2.3. **Abelian varieties.**

Definition 2.3.1 (Abelian variety over a field)**.** Let k be a field. An *abelian variety over k* is a complete, connected algebraic group variety defined over k, i.e.

- A is a smooth (Definition 1.2.3), proper (Definition 1.2.2), geometrically connected (Definition A.2.6) algebraic variety over k,
- A is endowed with a group structure defined by morphisms of varieties over k (multiplication $m : A \times_k A \to A$ and inverse $i : A \to A$),
- the group law satisfies the group axioms scheme-theoretically.

In particular, an abelian variety is a projective algebraic group variety (Definition 2.1.1) over k.

Definition 2.3.2 (Dual abelian variety)**.** Let A be an abelian variety (Definition 2.3.1) over a field k. The *dual abelian variety of A*, denoted by A^*, is the abelian variety representing the Picard functor $\mathrm{Pic}_{A/k}$, i.e., (♠ TODO: picard functor)

$$A^* = \mathrm{Pic}^0(A),$$

where $\mathrm{Pic}^0(A)$ is the connected component of the identity in the Picard scheme of A. For any commutative k-algebra R, the group $A^*(R)$ classifies line bundles on $A \times_k \operatorname{Spec} R$ that have degree zero on every fiber over $\operatorname{Spec} R$.

Definition 2.3.3. Let A and B be abelian varieties over a field k. We say that A and B are *isogenous*, often written $A \sim B$, if there exists an k-isogeny (Definition 4.0.2) $A \to B$. The relation $\sim$ turns out to be an equivalence relation on the set of abelian varieties over a fixed field k.

Definition 2.3.4 (Isogeny of abelian varieties over a field)**.** (♠ TODO: finite group scheme) Let k be a field (Definition A.1.3) and let A and B be abelian varieties (Definition 2.3.1) over k. An *isogeny of abelian varieties over k* is a homomorphism $\varphi\colon A \to B$ of group schemes over k that is surjective and has finite kernel (Definition 2.1.6). The kernel $\ker\varphi$ is a finite group scheme over k (Definition 2.1.6). Equivalently, φ is a finite, faithfully flat homomorphism of abelian varieties over k.

Definition 2.3.5 (Polarization of an abelian variety)**.** Let A be an abelian variety (Definition 2.3.1) over a field (Definition A.1.3) k. Let $A^* = \mathrm{Pic}^0(A)$ be the dual abelian variety (Definition 2.3.2) of A. For any line bundle L on A, define $\phi_L\colon A \to A^*$ by

$$\phi_L(x) = t_x^* L \otimes L^{-1},$$

where $t_x\colon A \to A$ is translation by x. The map ϕ_L is a homomorphism of abelian varieties. A *polarization on A* is an isogeny $\phi\colon A \to A^*$ such that there exists an ample line bundle L on A with $\phi = \phi_L$. When this holds, we say that (A, ϕ) is a *polarized abelian variety*.

Definition 2.3.6 (Principal polarization of an abelian variety)**.** Let (A, ϕ) be a polarized abelian variety (Definition 2.3.5) over a field (Definition A.1.3). The polarization ϕ is called *principal* if it is an isomorphism (equivalently, if the degree of ϕ, denoted by $\deg(\phi)$, equals 1). An abelian variety equipped with a principal polarization is called a *principally polarized abelian variety* (PPAV).

For $g = 1$, a PPAV is an elliptic curve. A principally polarized abelian surface ($g = 2$) or threefold ($g = 3$) is the Jacobian of a smooth connected algebraic curve unless it decomposes as a product together with the product polarization.

Definition 2.3.7 (Weil–Châtelet group of an abelian variety over a field)**.** Let k be a field, and let A be an abelian variety defined over k. Denote by $G_k = \mathrm{Gal}(\overline{k}/k)$ the absolute Galois group of k.

The *Weil–Châtelet group of A over k*, denoted $\mathrm{WC}(A/k)$, is defined as the collection equivalence classes of homogeneous spaces for A/k.

3. Picard groups and Jacobian varieties

Definition 3.0.1. (♠ TODO: picard group of an algebraic space) Let X be a scheme. The *Picard group of X*, denoted by $\mathrm{Pic}(X)$, is the group of isomorphism classes of invertible sheaves (Definition A.3.5) (line bundles) on X (with respect to the Zariski topology), with the group operation induced by the tensor product of sheaves (Definition A.3.6).

More precisely:

1. Two invertible sheaves (Definition A.3.1) $\mathcal{L}$ and $\mathcal{M}$ are isomorphic if there exists an $\mathcal{O}_X$-module (Definition A.3.4) isomorphism $\phi : \mathcal{L} \xrightarrow{\sim} \mathcal{M}$.
2. The tensor product (Definition A.3.6) of invertible sheaves induces a well-defined group operation on the set of isomorphism classes. (♠ TODO: sheafy dual)
3. The inverse of an invertible sheaf $\mathcal{L}$ is given by its dual $\mathcal{L}^\vee = \mathscr{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$ (Definition A.3.7).
4. The identity element is the class of the structure sheaf $\mathcal{O}_X$.

(♠ TODO: class group of ring) When $X = \mathrm{Spec}(A)$ is an affine scheme (Definition 1.2.4), there is a canonical isomorphism

$$\mathrm{Pic}(\mathrm{Spec}(A)) \cong \mathrm{Cl}(A),$$

where $\mathrm{Cl}(A)$ is the class group of the ring A (the group of isomorphism classes of rank-1 projective modules over A, also known as the Picard group of the ring).

(♠ TODO: extract this out) If X is a Noetherian, integral (Definition 1.2.5), and locally factorial scheme, then $\mathrm{Pic}(X)$ is isomorphic to the Weil divisor class group $\mathrm{Cl}(X)$.

3.1. The relative Picard functor of a scheme over a scheme.

Definition 3.1.1. Let X be a scheme over a scheme S. The fppf sheaf (Definition A.3.1) associated to the functor

$$P_{X/S} : (\mathrm{Sch}\,/S)^{\circ} \to \mathbf{Sets}, \quad T \mapsto \mathrm{Pic}(X \times_S T)$$

is denoted by $\mathrm{Pic}_{X/S}$ and is called the *relative Picard functor of X over S*.

For an S-scheme T, the group $\mathrm{Pic}_{X/S}(T)$ is called the *relative Picard group of $X \times_S T$ over T*.

Proposition 3.1.2 (see [BLR12, Chapter 8, after Definition 2 and after Proposition 3])**.** Let $f : X \to S$ be a scheme over a scheme. The relative Picard functor (Definition 3.1.1)

$$\mathrm{Pic}_{X/S} : (\mathrm{Sch}\,/S)^{\circ}_{\mathrm{fppf}} \to \mathbf{Sets}$$

can be constructed as

$$\mathrm{Pic}_{X/S}(T) = H^0(T, R^1 f_*(\mathbb{G}_m))$$

(Definition A.3.8).

If f is proper, then this formula remains valid when H^0 and $R^1 f_*(\mathbb{G}_m)$ are computed via the étale topology.

Proposition 3.1.3. Let X be a scheme. The sheaf cohomology groups

$$H^1(X, \mathbb{G}_m)$$

(Definition A.3.8) with respect to the (big) fpqc, fppf, étale, and Zariski topologies are all naturally isomorphic to one another and are isomorphic to the Picard group of X.

3.2. Representability of the relative Picard functor.

Theorem 3.2.1 ([Gro66, n°232, Théorème 3.1], see [BLR12, Section 8.2, Theorem 1])**.** (♠ TODO: projective morhpism of schemes) Let $f : X \to S$ be a projective, finitely presented, flat morphism of schemes (Definition 1.2.1) whose geometric fibers (Definition A.2.4) are integral (Definition 1.2.5). The fppf-sheaf (Definition A.3.1)

$$\mathrm{Pic}_{X/S} : (\mathrm{Sch}\,/S)^{\circ}_{\mathrm{fppf}} \to \mathbf{Sets}$$

(Definition 3.1.1) is representable by a separated S-scheme which is locally of finite presentation over S.

Theorem 3.2.2 (Mumford, unpublished, see [BLR12, Section 8.2, Theorem 2])**.** Let $f : X \longrightarrow S$ be flat, projective, and finitely presented with geometrically reduced fibres. Assume that the irreducible components of the fibres of f are geometrically irreducible.

The fppf-sheaf (Definition A.3.1)

$$\mathrm{Pic}_{X/S} : (\mathrm{Sch}\,/S)^{\circ}_{\mathrm{fppf}} \to \mathbf{Sets}$$

(Definition 3.1.1) is representable by by a (not necessarily separated) S-scheme which is locally of finite presentation over S.

Theorem 3.2.3 ([Mur64], [Oor62], see [BLR12, Section 8.2, Theorem 3])**.** Let X be a proper (Definition 1.2.2) scheme over a field (Definition A.1.3) k. The fppf-sheaf (Definition A.3.1)

$$\mathrm{Pic}_{X/k} : (\mathrm{Sch}\,/k)^{\circ}_{\mathrm{fppf}} \to \mathbf{Sets}$$

(Definition 3.1.1) is representable by by a locally of finite type scheme over k.

Proposition 3.2.4 ([BLR12, Section 8.4, Proposition 2])**.** Let $f : X \longrightarrow S$ be a proper (Definition 1.2.2), flat, locally of finite presentation morphism of schemes (Definition 1.2.1). Let s be a point of S such that $H^2(X_s, \mathcal{O}_{X_s}) = 0$. (♠ TODO: for what topology) Then there exists an open neighborhood U of s such that $\mathrm{Pic}_{X/S}\big|_U$ (Definition 3.1.1) is formally smooth over U. (♠ TODO: formally smooth) (♠ TODO: a theorem that says that sheaf cohomology for coherent sheaves on the zariski, etale, fppf topologies are all the same.)

Definition 3.2.5. Let S be an algebraic stack.

3.3. **Picard variety.**

Definition 3.3.1. Let X/k be a smooth, geometrically integral projective variety over a field (Definition A.1.3) k.

The relative Picard functor (Definition 3.1.1) $\mathrm{Pic}_{X/k}$ is representable by a smooth scheme (Definition 1.2.3) over k. (♠ TODO: show that the identity component is an abelian variety) The identity component (Definition 2.1.2) $\mathrm{Pic}^0_{X/k}$ is in fact an abelian variety and is called the *Picard variety of X/k*.

(♠ TODO: representability by algebraic space)

3.4. **Jacobian variety of a curve.**

Theorem 3.4.1. Let X be a proper (Definition 1.2.2) curve over a field (Definition A.1.3) k. The relative Picard functor (Definition 3.1.1) $\mathrm{Pic}_{X/k}$ is representable by a smooth scheme (Definition 1.2.3) over k.

Definition 3.4.2 (Jacobian variety of a curve)**.** (♠ TODO: separate out jacobian of just a proper curve and the jacobian of a nice curve) Let X be a proper (Definition 1.2.2) curve over a field (Definition A.1.3) k.

1. The relative Picard functor (Definition 3.1.1) $\mathrm{Pic}_{X/k}$ is representable by a smooth (Definition 1.2.3) group scheme (Definition 2.1.1) over k. The *(generalized) Jacobian of X* refers to the identity component (Definition 2.1.2) $\mathrm{Pic}^0_{X/k}$ of this group scheme.
2. Now assume that X is smooth, is projective, and has geometrically connected (Definition A.2.6) connected components. The Jacobian is an abelian variety (Definition 2.3.1) over k and may be called the *Jacobian variety of X*. It may be denoted by notations such as $\mathrm{Jac}(X)$, Jac_X, or J_X.

   Under these assumptions, note that $\mathrm{Jac}(X)$ coincides with the Picard variety (Definition 3.3.1) of X/k. Equivalently, J_X parametrizes isomorphism classes of degree-zero line bundles (Definition A.3.5) on X, where a line bundle $\mathcal{L}$ has *degree zero* if its degree (computed as the sum of orders of any rational function defining it) vanishes.

The Jacobian variety $\mathrm{Jac}(X)$ has dimension g where g is the geometric genus of X.

When X is equipped with a base point, the Jacobian variety carries a canonical principal polarization (Definition 2.3.6) Θ, making (J_X, Θ) a principally polarized abelian variety.

Proposition 3.4.3 (see e.g. [Mil08, Chapter III, Proposition 2.1.]). (♠ TODO: tangent space) Let X be a smooth (Definition 1.2.3), projective curve whose connected components are geometrically connected (Definition A.2.6). The tangent space to the Jacobian variety (Definition 3.4.2) $\mathrm{Jac}(X)$ at the identity point is canonically isomorphic to $H^1(X, \mathcal{O}_X)$. Consequently, the dimension of $\mathrm{Jac}(X)$ equals the geometric genus of X.

3.5. **Coverings of curves from the Jacobian.**

Proposition 3.5.1 (see e.g. [Mil08, Chapter III, Proposition 9.1.]). Let X be a smooth (Definition 1.2.3), projective curve over a field (Definition A.1.3) k whose connected components are geometrically connected (Definition A.2.6).

(♠ TODO: etale covering)

1. Suppose that $J' \to \mathrm{Jac}(X)$ is a connected étale covering of the Jacobian variety (Definition 3.4.2) $\mathrm{Jac}(X)$. The pullback $X \times_{\mathrm{Jac}(X)} J' \to X$ is a connected étale covering of X.
2. Every connected abelian étale covering of X is obtained via connected étale coverings of $\mathrm{Jac}(X)$ as above.
   (♠ TODO: do I need to assume the curve to be connected) (♠ TODO: define abelianization)
3. Let $P \in X(k)$. There is a map $\pi_1^{\text{ét}}(X, P)^{\text{ab}} \to \pi_1^{\text{ét}}(\mathrm{Jac}(X), 0)$ **??** induced by this choice of P and it is an isomorphism where 0 is the identity point of $\mathrm{Jac}(X)$.

Proposition 3.5.2. (♠ TODO: talk about how

$$H^1(X_{\text{et}}, G) \cong \mathrm{Hom}_{\text{cont}}(\pi_1^{\text{et}}(X, \overline{x}), G)$$

for finite groups G and general connected schemes X.)

Proposition 3.5.3 (see [Mil08, Chapter III, Corollary 9.6.]). Let X be a smooth (Definition 1.2.3), projective curve over a field (Definition A.1.3) k whose connected components are geometrically connected (Definition A.2.6). (♠ TODO: handle the disconnectedness of X; apparently the abel-jacobi map would need to pick out rational points on all components.) Let $P \in X(k)$. For all primes ℓ, the map of étale cohomologyg roups $H^1(\mathrm{Jac}(X), \mathbb{Z}_\ell) \to H^1(X, \mathbb{Z}_\ell)$ induced by the embedding $X \hookrightarrow \mathrm{Jac}(X)$ is an isomorphism. (♠ TODO: apparently this is still true for $\ell = p$ by a Witt vector arguement)

(♠ TODO: look into SGA 1, Expose XI for result that relates etale fundamental group of abelian variety to adic tate module)

4. Abelian schemes

Definition 4.0.1 (Abelian scheme)**.** Let S be a scheme. An *abelian scheme over S* is a proper (Definition 1.2.2) and smooth (Definition 1.2.3) group scheme (Definition 2.1.1)

$$\pi : A \to S$$

with geometrically connected (Definition A.2.6) fibers. Each geometric fiber (Definition A.2.4) $A_{\overline{s}}$ (over a geometric point $\overline{s} \to S$) is then an abelian variety (Definition 2.3.1).

In particular, over a field, an abelian scheme is precisely an abelian variety.

Definition 4.0.2 (Isogeny of abelian schemes)**.** Let S be a scheme, and let A and B be abelian schemes over S (Definition 4.0.1). An *isogeny of abelian schemes over S* or *S-isogeny of abelian schemes* is a morphism of group schemes

$$\varphi : A \to B$$

(♠ TODO: define finite, faithfully flat, and sujrective scheme morpshisms) over S which is finite, faithfully flat, and surjective.

Equivalently, φ is a morphism of abelian schemes whose geometric fibers (Definition A.2.4) are *isogenies of abelian varieties*, i.e., surjective homomorphisms with finite kernel.

The kernel of φ is often denoted by $\ker \varphi$ or $A[\varphi]$.

5. Néron models

Definition 5.0.1 (Néron model of an abelian variety)**.** Let S be an integral scheme (Definition 1.2.5), let η be its generic point (Definition 1.1.4), and let $K = \mathcal{O}_{S,\eta}$ be the function field (Definition A.2.3) of S. Let A_K be an abelian variety over K (Definition 2.3.1).

A *Néron model of A_K over S* is a smooth (Definition 1.2.3) separated group scheme $\mathcal{A}$ of finite type over S whose generic fiber (Definition 1.1.3) $\mathcal{A}_\eta$ is isomorphic to A_K, satisfying the following universal property (the *Néron mapping property*): for every smooth S-scheme U and every K-morphism $f_K : U_\eta \to A_K$, there exists a unique S-morphism $f : U \to \mathcal{A}$ extending f_K.

If such a model exists, it is unique up to unique isomorphism.

6. Reduction of abelian varieties

6.1. Good and semi-abelian reduction.

Definition 6.1.1 (Good reduction of an abelian variety over a Dedekind scheme)**.** Let S be a Dedekind scheme (Definition 1.1.1), let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S. Let A_K be an abelian variety over K (Definition 2.3.1), and let $s \in S$ be a closed point.

We say that A_K has *good reduction at s* if there exists a smooth and proper scheme $\mathcal{A}$ over S whose generic fiber $\mathcal{A}_\eta$ is isomorphic to A_K. In this case, the special fiber $\mathcal{A}_s = \mathcal{A} \times_S \operatorname{Spec} k(s)$ (where $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$ is the residue field at s) is an abelian variety over $k(s)$, and we say that $\mathcal{A}_s$ is the *good reduction of A_K at s*.

Equivalently, A_K has good reduction at s (Definition 6.1.1) if and only if there is a Néron model (Definition 5.0.1) $\mathcal{A}$ of A over the local ring $\mathcal{O}_{S,s}$ (or its completion $\widehat{\mathcal{O}}_{S,s}$ is proper (Definition 1.2.2) (or equivalently an abelian scheme (Definition 4.0.1)).

Definition 6.1.2 (Abelian and semi-abelian reduction for a smooth group scheme over a Dedekind scheme)**.** Let S be a Dedekind scheme (Definition 1.1.1), let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S. Assume for simplicity that S is connected.

Let G be a smooth (Definition 1.2.3) group scheme of finite type (Definition 2.1.1) over S. For any closed point $s \in S$, the special fiber of G at s is the group scheme $G_s = G \times_S \operatorname{Spec} k(s)$ over the residue field $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$. Let $(G_s)^0$ denote the identity component (Definition 2.1.2) of G_s.

We say that G has *abelian reduction at s* if $(G_s)^0$ is an abelian variety over $k(s)$ (Definition 2.3.1). We say that G has *semi-abelian reduction at s* if $(G_s)^0$ fits into a short exact sequence of algebraic groups over $k(s)$:

$$1 \to T \to (G_s)^0 \to B \to 0,$$

where T is an algebraic torus over $k(s)$ (Definition 2.2.1) and B is an abelian variety over $k(s)$.

In particular, if G is a Néron model (Definition 5.0.1) of its generic fiber G_K, we say that G_K has *abelian (resp. semi-abelian) reduction at s* if the corresponding fact holds for G. This is equivalent to saying that the base change $G \times_S \operatorname{Spec} \mathcal{O}_{S,s}$ of G to $\operatorname{Spec} \mathcal{O}_{S,s}$ has abelian (resp. semi-abelian) reduction at the unique closed point of $\operatorname{Spec} \mathcal{O}_{S,s}$.

6.2. **Potential reduction.**

Definition 6.2.1 (Potential good and semi-abelian reduction for an abelian variety over the function field of a Dedekind scheme)**.** Let S be a Dedekind scheme (Definition 1.1.1), let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S. Let A_K be an abelian variety over K (Definition 2.3.1), and let $s \in S$ be a closed point.

We say that A_K has *potential good reduction at s* (resp. *potential abelian reduction at s*, resp. *potential semi-abelian reduction at s*) if there exists a finite Galois extension (Definition A.1.4) L of K such that $A_L = A_K \times_K L$ has abelian (resp. semi-abelian) reduction at all closed points $s' \in S'$ lying over s, where S' is the normalization of S in L. Instead of potential abelian reduction, we will also talk about *potential good reduction*.

7. Galois representations and Tate modules

7.1. **Absolute Galois groups and decomposition theory.**

Definition 7.1.1 (Separable closure of a field)**.** Let K be a field (Definition A.1.3). A *separable closure of K* is a separable algebraic extension K^{sep} of K such that every element of K^{sep} is separable over K, and K^{sep} has no nontrivial separable algebraic extensions (i.e., the only separable algebraic extension of K^{sep} inside a fixed algebraic closure $\overline{K}$ is itself).

K^{sep} is often also denoted as K^s, K_s, or $\overline{K}$, the latter of which should not be mistaken for notation for an algebraic closure of K.

(♠ TODO: compositum of fields) Equivalently, K^{sep} is the compositum of all finite (Definition A.1.7) separable subextensions of K in a fixed algebraic closure $\overline{K}$. A separable closure always exists and is unique up to (non-unique) isomorphism over K. If K has characteristic (Definition A.1.2) zero, then every algebraic extension is separable, so the separable closure coincides with the algebraic closure.

Definition 7.1.2 (Absolute Galois group of a field)**.** Let K be a field (Definition A.1.3), and let K^s be a separable closure (Definition 7.1.1) of K. Write $\overline{K}$ for an algebraic closure of K containing K^s.

The *absolute Galois group of K* is the Galois group (Definition A.1.4):

$$G_K = \mathrm{Gal}(K^s/K)$$

equipped with the Krull topology (Definition A.1.5), making it a profinite group. Equivalently, G_K is the inverse limit (Definition A.1.6) of the finite Galois groups $\mathrm{Gal}(L/K)$ as L ranges over all finite (Definition A.1.7) Galois subextensions of K^s/K:

$$G_K = \varprojlim_{L} \mathrm{Gal}(L/K)$$

We note that $\overline{Aut}(\overline{K}/K)$ is naturally isomorphic as a profinite group to the absolute Galois group $\mathrm{Gal}(K^s/K)$. One often writes $\mathrm{Gal}(\overline{K}/K)$ for $\overline{Aut}(\overline{K}/K)$, even when $\overline{K}/K$ is not a Galois extension (which is the case when K is not a perfect field). On the other hand, if K is a perfect field, then the separable closure K^s coincides with the algebraic closure $\overline{K}$, and $\overline{K}/K$ is a Galois extension, so the notation of $\mathrm{Gal}(\overline{K}/K)$ is no longer an abuse of notation.

Definition 7.1.3 (Decomposition group of a Dedekind scheme at a closed point)**.** Let S be a Dedekind scheme (Definition 1.1.1) with function field K. Fix a separable closure (Definition 7.1.1) $\bar{K}$ of K, and let $G_K = \mathrm{Gal}(\bar{K}/K)$ be the absolute Galois group (Definition 7.1.2) of K. Let $s \in S$ be a closed point (Definition 1.1.5).

Choosing a prime $\bar{\mathfrak{s}}$ of the integral closure (Definition A.2.2) of $\mathcal{O}_{S,s}$ in $\bar{K}$ is equivalent to choosing a geometric point $\bar{s} \to S$ lying over s. The *decomposition group of S at s (with respect to this choice)* is the subgroup:

$$D_s = D(\bar{\mathfrak{s}}|s) = \{\sigma \in G_K \mid \sigma(\bar{\mathfrak{s}}) = \bar{\mathfrak{s}}\} \subset G_K$$

The decomposition group D_s is well-defined up to conjugacy in G_K. There is a natural surjection $\phi : D_s \twoheadrightarrow \mathrm{Gal}(\overline{k(s)}/k(s))$, where $\overline{k(s)}$ is a separable closure of the residue field $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$. The kernel of this surjection is the inertia subgroup.

Definition 7.1.4 (Inertia group in the absolute Galois group of the function field of a Dedekind scheme at a closed point)**.** Let S be a Dedekind scheme (Definition 1.1.1) with function field K. Fix a separable closure (Definition 7.1.1) $\bar{K}$ of K, and let $G_K = \mathrm{Gal}(\bar{K}/K)$ be the absolute Galois group (Definition 7.1.2) of K. Let $s \in S$ be a closed point (Definition 1.1.5) with residue field (Definition 1.2.9) $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$.

Choosing a prime $\bar{\mathfrak{s}}$ of the integral closure of $\mathcal{O}_{S,s}$ in $\bar{K}$ is equivalent to choosing a geometric point $\bar{s} \to S$ lying over s. This choice determines the decomposition group (Definition 7.1.3) $D_s = D(\bar{\mathfrak{s}}|s) \subset G_K$, together with a natural surjection $\phi : D_s \twoheadrightarrow \mathrm{Gal}(\overline{k(s)}/k(s))$, where $\overline{k(s)}$ is a separable closure of $k(s)$.

The *inertia group in the absolute Galois group of the function field of S at s (with respect to the choice of $\bar{\mathfrak{s}}$)* is the subgroup:

$$I_s = I(\bar{\mathfrak{s}}|s) = \ker\left(D_s \xrightarrow{\phi} \mathrm{Gal}(\overline{k(s)}/k(s))\right) \subset G_K$$

Equivalently, I_s consists of all automorphisms $\sigma \in G_K$ that induce the identity on the residue field (Definition 1.2.9) of the valuation ring (Definition A.2.1) corresponding to $\bar{\mathfrak{s}}$.

7.2. Tate modules and unramified representations.

Definition 7.2.1 (Tate module of an abelian variety)**.** (♠ TODO: separable closure) Let K be a field (Definition A.1.3), let $\bar{K}$ be a separable closure of K, and let $G_K = \mathrm{Gal}(\bar{K}/K)$ be the absolute Galois group (Definition A.1.4) of K. Let A_K be an abelian variety (Definition 2.3.1) over K, and let ℓ be a prime number (Definition A.1.1) that is distinct from the characteristic (Definition A.1.2) of K.

For any integer $\nu \geq 1$, let $A_K[\ell^\nu]$ be ℓ^ν-torsion subgroup of A_K, which is a finite étale group scheme over K (or equivalently, as a discrete G_K-module via its $\bar{K}$-rational points). The *ℓ-adic Tate module of A_K* is the G_K-module:

$$T_\ell(A_K) = \varprojlim_{\nu} A_K[\ell^\nu](\bar{K})$$

where the inverse limit is taken with respect to the multiplication-by-ℓ transition maps. One may also consider the rational Tate module:

$$V_\ell(A_K) = T_\ell(A_K) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

which is a vector space over $\mathbb{Q}_\ell$ of dimension equal to the relative dimension of A_K, equipped with a continuous linear action of G_K.

Definition 7.2.2 (Unramified Galois representation at a closed point)**.** Let S be a Dedekind scheme (Definition 1.1.1) with function field (Definition A.2.3) K. Fix a separable closure (Definition 7.1.1) $\bar{K}$ of K, and let $G_K = \mathrm{Gal}(\bar{K}/K)$ be the absolute Galois group (Definition 7.1.2) of K. Let $s \in S$ be a closed point (Definition 1.1.5), and fix a prime $\bar{\mathfrak{s}}$ of the integral closure (Definition A.2.2) of $\mathcal{O}_{S,s}$ in $\bar{K}$. Let $I_s \subset G_K$ be the corresponding inertia subgroup (Definition 7.1.4).

Let M be a continuous (Definition A.4.2) G_K-module (such as a finite torsion module or an ℓ-adic Tate module (Definition 7.2.1)). We say that M is *unramified at s* if the inertia group (Definition 7.1.4) I_s acts trivially (Definition A.4.1) on M, i.e.,

$$\sigma \cdot m = m \quad \text{for all } \sigma \in I_s \text{ and all } m \in M.$$

In particular, for an abelian variety A_K over K and a prime ℓ distinct from the characteristic of $k(s)$, we say that the Tate module $T_\ell(A_K)$ (resp. $V_\ell(A_K)$) is *unramified at s* if I_s acts trivially on $T_\ell(A_K)(\bar{K})$ (resp. $V_\ell(A_K)$).

Remark 7.2.3. Although the inertia group I_s depends on the choice of $\bar{\mathfrak{s}}$, the property of being unramified at s does not: all inertia groups at s are conjugate in G_K, and since any continuous G_K-module is stable under the full Galois group, if one inertia subgroup acts trivially on M, then all do.

8. The criterion of Néron–Ogg–Shafarevich

Theorem 8.0.1 (Criterion of Néron–Ogg–Shafarevich for good reduction of abelian varieties)**.** (♠ TODO: generic point) Let S be a Dedekind scheme (Definition 1.1.1), let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field (Definition A.2.3) of S. Let A_K be an abelian variety over K (Definition 2.3.1), and let $s \in S$ be a closed point (Definition 1.1.5). Fix a prime $\bar{\mathfrak{s}}$ of the integral closure (Definition A.2.2) of $\mathcal{O}_{S,s}$ in a separable closure (Definition 7.1.1) $\bar{K}$ of K, and let $I_s \subset G_K = \mathrm{Gal}(\bar{K}/K)$ be the corresponding inertia subgroup (Definition 7.1.4). Let ℓ be a prime number (Definition A.1.1) distinct from the characteristic (Definition A.1.2) of the residue field (Definition 1.2.9) $k(s)$.

The following conditions are equivalent:

1. A_K has good reduction at s (Definition 6.1.1).
2. There exists an abelian scheme (Definition 4.0.1) $\mathcal{A}$ over S whose generic fiber (Definition 1.1.3) is isomorphic to A_K. In this case, the Néron model (Definition 5.0.1) of A_K over S is an abelian scheme over S, and its special fiber $\mathcal{A}_s$ is an abelian variety over $k(s)$.
3. The inertia group I_s acts trivially on the ℓ^ν-torsion points $A_K[\ell^\nu](\bar{K})$ for some integer $\nu \geq 1$.
4. The inertia group I_s acts trivially on the full ℓ-adic Tate module (Definition 7.2.1) $T_\ell(A_K)(\bar{K})$.
5. The inertia group I_s acts trivially on the rational ℓ-adic Tate module $V_\ell(A_K) = T_\ell(A_K) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.
6. The Tate module $T_\ell(A_K)$ (equivalently, $V_\ell(A_K)$) is an unramified (Definition 7.2.2) continuous G_K-representation at s.

Moreover, if A_K has good reduction (Definition 6.1.1) at s with abelian scheme model $\mathcal{A}$, then for every prime ℓ' distinct from $\mathrm{char}(k(s))$, the Tate module $T_{\ell'}(A_K)$ is unramified at s. Furthermore, for any such ℓ', the specialization map induces an isomorphism of $G_{k(s)}$-modules:

$$T_{\ell'}(A_K) \xrightarrow{\sim} T_{\ell'}(\mathcal{A}_s)$$

where the $G_{k(s)} = \mathrm{Gal}(\overline{k(s)}/k(s))$-action on the left is via the identification $G_{k(s)} \cong D_s/I_s$.

9. Selmer theory

Definition 9.0.1. Assume **??** and further assume that L/K is a finite (Definition A.1.7) Galois extension (Definition A.1.4).

The *decomposition group of L/K* is the subgroup

$$D_w = \{\sigma \in \mathrm{Gal}(L/K) : w \circ \sigma = w\}$$

(Definition A.1.4)

Definition 9.0.2. Assume **??** and further assume that L/K is a finite (Definition A.1.7) Galois extension (Definition A.1.4).

There is a natural surjective homomorphism

$$D_w \to \mathrm{Gal}(k_w/k_v)$$

on the residue fields. The *inertia group of L/K* is its kernel:

$$I_w = \{\sigma \in D_w : \sigma(x) \equiv x \pmod{\mathfrak{m}_w} \text{ for all } x \in \mathcal{O}_w\}$$

(Definition 9.0.1)

Definition 9.0.3. Assume **??** and further assume that L/K is a Galois extension (Definition A.1.4).

The *decomposition group of L/K* is the subgroup

$$D_w = \{\sigma \in \mathrm{Gal}(L/K) : w \circ \sigma = w\}.$$

(Definition A.1.4) It is also the inverse limit (Definition A.1.6) of the decomposition groups (Definition 9.0.1) of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of $\mathrm{Gal}(L/K)$.

Letting (K, v) be a valuation field, its *decomposition group* $D_{K,v}$ (also sometimes denoted by $G_{K,v}$) is the decomposition group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Definition 9.0.4. Assume **??** and further assume that L/K is a Galois extension (Definition A.1.4).

There is a natural surjective homomorphism

$$D_w \to \mathrm{Gal}(k_w/k_v)$$

(Definition 9.0.3) on the residue fields. The *inertia group of L/K* is its kernel:

$$I_w = \{\sigma \in D_w : \sigma(x) \equiv x \pmod{\mathfrak{m}_v} \text{ for all } x \in \mathcal{O}_v\}.$$

(Definition 9.0.3) (Definition A.1.4) It is also the inverse limit (Definition A.1.6) of the decomposition groups (Definition 9.0.1) of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of D_w.

Letting (K, v) be a valuation field, its *inertia group* $I_{K,v}$ is the inertia group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Definition 9.0.5 (Kummer map associated to an isogeny of an abelian variety)**.** Let k be a field (Definition A.1.3), and let A and B be commutative group schemes (Definition 2.1.1) over S. Suppose $\varphi : A \to B$ is an isogeny over S. Denote by $G_k = \text{Gal}(\overline{k}/k)$ the absolute Galois group (Definition 7.1.2) of k.

The short exact sequence

$$0 \to \ker \varphi \to A(\overline{k}) \xrightarrow{\varphi} B(\overline{k}) \to 0.$$

yields the following long exact sequence in Galois cohomology:

$$\begin{aligned} 0 \to \ker \varphi(k) \to & A(k) \xrightarrow{\varphi} B(k) \\ \xrightarrow{\delta_\varphi} H^1(G_k, \ker \varphi) \to & H^1(G_k, A(\overline{k})) \to H^1(G_k, B(\overline{k})) \to \cdots . \end{aligned}$$

The *Kummer map associated to the isogeny φ* is the Galois cohomological connecting homomorphism

$$\delta_\varphi : B(k) \to H^1(G_k, \ker \varphi),$$

above. In fact, there is a short exact sequence

$$0 \to B(k)/\varphi(A(k)) \xrightarrow{\delta'_\varphi} H^1(G_k, \ker \varphi) \to H^1(G_k, A)[\varphi] \to 0$$

where the map δ'_φ is induced by δ_φ. We may also let the *Kummer map associated to φ* refer to this map δ'_φ. We may also use the abuse of notation δ_φ to denote δ'_φ.

Definition 9.0.6 (Selmer group for an isogeny of abelian group schemes over a global field)**.** (♠ TODO: It should be possible to define this for more general group schemes) Let K be a global field, and let A and B be abelian group schemes (Definition 2.1.1) defined over K. Suppose $\varphi : A \to B$ is an isogeny defined over K.

Denote by $G_K = \text{Gal}(\overline{K}/K)$ the absolute Galois group (Definition 7.1.2) of K. For each place v of K, fix an extension v to $\overline{K}$, which yields an embedding $\overline{K} \subset \overline{K}_v$ and a decomposition group (Definition 9.0.3) $G_v \subset G_K$.

The *Selmer group of φ over K*, denoted by notations such as $\text{Sel}(A/K)$, $\text{Sel}^\varphi(A/K)$, $\text{Sel}^{(\varphi)}(A/K)$, $\text{Sel}_\varphi(A/K)$, is the subgroup of the Galois cohomology group $H^1(K, \ker \varphi)$ given by

$$\text{Sel}^\varphi(A/K) := \ker \left(H^1(G_K, A[\varphi]) \to \prod_v H^1(G_v, A(\overline{K}))[\varphi] \right),$$

(♠ TODO: define the kummer map associated to φ) where the product runs over all places v of K, and $H^1(G_v, A)[\varphi]$ denotes the image of the local Kummer map associated to φ.

We describe the map

$$\text{(A)} \qquad H^1(G_K, A[\varphi]) \to \prod_v H^1(G_v, A(\overline{K}))[\varphi]$$

(♠ TODO: define a decomposition group) used to define the kernel above: Note that G_v acts on $A(\overline{K}_v)$ and $B(\overline{K}_v)$, and the base change φ_v of φ to K_v induces a Kummer short exact sequence (Definition 9.0.5)

$$0 \to B(K_v)/\varphi(A(K_v)) \xrightarrow{\delta_{\varphi_v}} H^1(G_v, A[\varphi]) \to H^1(G_v, A(\overline{K}_v))[\varphi] \to 0.$$

The natural inclusions $G_v \subset G_K$ and $E(K) \subset E(K_v)$ give restriction maps $H^1(G_K, A[\varphi]) \to H^1(G_v, A[\varphi])$ on Galois cohomology, and we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & B(K)/\varphi(A(K)) \arrow[r, "\delta_\varphi"] \arrow[d] & H^1(G_K, A[\varphi]) \arrow[r] \arrow[d, "\prod_v \mathrm{res}_v"] \arrow[dr, dotted] & H^1(G_K, A(\overline{K}))[\phi] \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \prod_v B(K_v)/\varphi(A(K_v)) \arrow[r, "\delta_{\varphi_v}"] & \prod_v H^1(G_v, A[\varphi]) \arrow[r] & \prod_v H^1(G_v, A(\overline{K}_v))[\phi] \arrow[r] & 0
\end{tikzcd}$$

The map (A) is the one given in the above commutative diagram by the dotted arrow.

Equivalently,

$$\mathrm{Sel}^{\varphi}(A/K) = \left\{c \in H^1(K, A[\varphi]) : \text{ for all places } v \text{ of } K, \mathrm{res}_v(c) \in \operatorname{Im} \delta_{\varphi_v}\right\}$$

(Proposition 9.0.7)

Proposition 9.0.7. Let K be a global field, and let A and B be abelian group schemes (Definition 2.1.1) defined over K. Let $\varphi : A \to B$ be an isogeny defined over K. Denote by $G_K = \mathrm{Gal}(\overline{K}/K)$ the absolute Galois group (Definition 7.1.2) of K. For each place v of K, fix an extension v to $\overline{K}$, which yields an embedding $\overline{K} \subset \overline{K}_v$ and a decomposition group (Definition 9.0.3) $G_v \subset G_K$. Write $\mathrm{res}_v : H^1(K, A[\varphi]) \to H^1(G_v, A[\varphi])$ for the restriction maps induced by the inclusions $G_v \subset G_K$.

The Selmer group (Definition 9.0.6) $\mathrm{Sel}(A/K)$ coincides with the following group:

$$\left\{c \in H^1(K, A[\varphi]) : \text{ for all places } v \text{ of } K, \mathrm{res}_v(c) \in \mathrm{Image}\left(\frac{B(K_v)}{\varphi(A(K_v))} \xrightarrow{\delta_{\varphi_v}} H^1(G_v, A[\varphi])\right)\right\}$$

(**??**) (**??**)

Proof. By the exactness of

$$0 \longrightarrow \prod_v B(K_v)/\varphi(A(K_v)) \xrightarrow{\delta_{\varphi_v}} \prod_v H^1(G_v, A[\varphi]) \xrightarrow{\pi} \prod_v H^1(G_v, A(\overline{K}_v))[\phi] \longrightarrow 0,$$

the image of δ_{φ_v} equals the kernel of π. By definition, the Selmer group is the kernel of the composition of π with

$$H^1(G_K, A[\varphi]) \to \prod_v H^1(G_v, A[\varphi])$$

and hence the Selmer group is contained in the kernel of π. Inversely, if the $c \in H^1(G_K, A[\varphi])$ is not an element of the Selmer group, then it is by definition sent to a nonzero element of

$\prod_v H^1(G_v, A(\overline{K}_v))[\phi]$, which implies that its image under $\prod_v \mathrm{res}_v$ cannot be an element of $\ker \pi = \mathrm{Im}\, \delta_{\varphi_v}$. □

Proposition 9.0.8 ([Če16, Proposition 2.7(d)]). (♠ TODO: figure out how to have local tamagawa numbers applicable to this setting) Let $S = \mathrm{Spec}\, \mathfrak{o}$ (Definition 1.2.4) where $\mathfrak{o}$ is a Henselian discrete valuation ring with fraction field K and residue field k. Let $\varphi : A \to B$ be a K-isogeny (Definition 2.3.4) of abelian varieties (Definition 2.3.1). Write $\delta_\varphi : B(K) \to H^1(K, A[\varphi])$ (Definition 2.1.6) for the Kummer (Definition 9.0.5) map associated to φ. Let $\mathcal{A}/S$ and $\mathcal{B}/S$ be the Néron models (Definition 5.0.1) of A and B respectively, and lift $\varphi : A \to B$ to an isogeny $\varphi : \mathcal{A} \to \mathcal{B}$. Further write $\varphi : (\mathcal{A}_k/\mathcal{A}_k^0) \to (\mathcal{B}_k/\mathcal{B}_k^0)$ (Definition 2.1.2) for the induced homomorphism between the étale k-group schemes of connected components of the fibers $\mathcal{A}_k$ and $\mathcal{B}_k$. Let c_A and c_B respectively be the local Tamagawa numbers of A and B.

Assuming that

1. $\mathcal{A}[\varphi] \to S$ is étale (which holds whenever $\mathrm{Char}\, k \nmid \deg \varphi$ for example (♠ TODO: ref)), and
2. We have
$$\#(\mathcal{A}_k/\mathcal{A}_k^0(k)) = \#(\varphi(\mathcal{A}_k/\mathcal{A}_k^0)(k)) = \#(\mathcal{B}_k/\mathcal{B}_k^0(k))$$
(which holds whenever $\deg \varphi$ is coprime to $c_A \cdot c_B$ (♠ TODO: ref))

we have
$$H^1_{\mathrm{unr}}(K, A[\varphi]) = \mathrm{Im}\, \delta_\varphi = H^1_{\mathrm{fppf}}(S, \mathcal{A}[\varphi])$$
(**??**) (**??**) (**??**) (♠ TODO: flat cohomology)

Corollary 9.0.9. (♠ TODO: if selmer groups can be extended to more general bases, then this should be generalizable to more general bases whose residue fields are all of a single characteristic) Let K be a global function field over a finite field $\mathbb{F}_q$, and let A and B be abelian group schemes (Definition 2.1.1) defined over K. Let $\varphi : A \to B$ be an isogeny defined over K. Denote by $G_K = \mathrm{Gal}(\overline{K}/K)$ the absolute Galois group (Definition 7.1.2) of K. For each place v of K, fix an extension v to $\overline{K}$, which yields an embedding $\overline{K} \subset \overline{K}_v$ and a decomposition group (Definition 9.0.3) $G_v \subset G_K$. Write $\mathrm{res}_v : H^1(K, A[\varphi]) \to H^1(G_v, A[\varphi])$ for the restriction maps induced by the inclusions $G_v \subset G_K$. Write $\delta_{\varphi_v} : B(K_v)/\varphi(A(K_v)) \to H^1(G_v, A[\varphi])$ for the local Kummer map (Definition 9.0.5) for A/K_v, and write $c_{A,v}$ and $c_{B,v}$ for the local Tamagawa numbers of A and B over the completion K_v of K at v.

If $\deg \varphi$ is coprime to $p \cdot \prod_v c_{A,v} \cdot \prod_v c_{B,v}$, then
$$\mathrm{Sel}^\varphi = \{c \in H^1(K, A[\varphi]) : \text{ for every place } v \text{ of } K, \mathrm{res}_v(C) \in H^1_{\mathrm{unr}}(K_p, A[\varphi])\}$$
(**??**)

Proof. Let S be the smooth projective curve over $\mathbb{F}_q$ corresponding to K. It is a Dedekind scheme of finite type over $\mathbb{F}_q$, so Néron models $\mathcal{A}$ and $\mathcal{B}$ of A and B over S exist. Writing $\varphi : \mathcal{A} \to \mathcal{B}$ for the isogeny lifting φ, we note that $\mathcal{A}[\varphi]$ is étale over S as a consequence of **??**. This holds by a combination of Proposition 9.0.7 and Proposition 9.0.8 — letting $\mathcal{A}$ and $\mathcal{B}$

be the Néron models of A and B over the smooth projective curve S over $\mathbb{F}_q$ corresponding to K $\square$

Appendix A. Background definitions

A.1. **Basic algebra and set theory.**

Definition A.1.1. A *prime number* is a positive element of $\mathbb{Z}$ that is an irreducible element of $\mathbb{Z}$.

Explicitly, a prime number p is a positive integer that is not 1 and whose only positive integer divisors are 1 and p

Equivalently, a prime number is a positive element of $\mathbb{Z}$ that is a prime element of $\mathbb{Z}$.

Definition A.1.2. Let R be a ring. There exists a unique ring homomorphism $\psi : \mathbb{Z} \to R$ defined by mapping $n \mapsto n \cdot 1_K$. The non-negative generator of the kernel of this map, $\ker(\psi) = \langle p \rangle \subseteq \mathbb{Z}$, is called the *characteristic of R*, denoted by $\mathrm{char}(R)$.

Definition A.1.3 (Field)**.** A *field* is commutative division ring. In other words, a field is a commutative ring for which all nonzero elements have a multiplicative inverse.

Definition A.1.4 (Galois Extension)**.** An extension L/K is called a *Galois extension* if it is both a normal extension and a separable algebraic extension. Its *Galois group*, usually denoted by $\mathrm{Gal}(L/K)$, is defined to be the automorphism group $\mathrm{Aut}(L/K)$.

Definition A.1.5. A *profinite set* is a set which is the projective limit (Definition A.1.6) of a (small) directed system of finite sets.

A profinite set $X = \varprojlim_i X_i$ has a natural topology, referred to as the *profinite topology*, *inverse limit topology*, or *Krull topology*. It can be formulated in the following equivalent ways:

(♠ TODO: define inverse limit topology)

1. As the inverse limit (Definition A.1.6) topology with respect to the family of projection maps $\pi_i : X \to X_i$ onto the finite sets X_i, each equipped with the discrete topology, in the system.
2. As the subspace topology on $\varprojlim_i X_i$ inherited from the product topology on $\prod_i X_i$.
3. As the topology generated by the basis of "finite index" sets (specifically, the clopen sets) of the form $\pi_i^{-1}(V)$ for some index i and some subset $V \subseteq X_i$.

These topologies do not depend on the inverse system representing X.

Definition A.1.6 (Special cases of limits)**.** Let $\mathcal{C}$ be a (large) category. Let I be a (large) category. Let $I \to \mathcal{C}$ be a diagram/system.

- Suppose that the system is a cofiltered system, i.e. I is a cofiltered category. A limit of this diagram is often denoted by

$$\varprojlim_{i \in I} D(i)$$

and may be called a *cofiltered (inverse/projective) limit*. In case that the system is more specifically an inverse/projective system, i.e. I is a cofiltered poset, the preferred term for such a limit is *inverse/projective limit*.
- Suppose that the system is a filtered system, i.e. I is a filtered category. A colimit of this diagram is often denoted by

$$\varinjlim_{i \in I} D(i)$$

and may be called a *filtered colimit* or a *direct/inductive/injective limit*. In case that the system is more specifically a direct/inductive system, i.e. I is a filtered poset, the preferred term for such a limit is *direct/inductive limit*.

Definition A.1.7 (Degree of an Extension, Finite Extension)**.** (♠ TODO: dimension) Let L/K be a field extension. The *degree* of the extension $[L : K] := \dim_K(L)$, i.e. the dimension of L as a K-vector space. If $[L : K] < \infty$, the extension L/K is called a *finite field extension* and L is said to be a *finite extension of K*.

A.2. **Schemes and topology.**

Definition A.2.1 (Valuation Ring of a Valued Field)**.** Let K be a field (Definition A.1.3) equipped with a valuation (♠ TODO: totally ordered abelian group) $v : K \to \Gamma$, where Γ is a totally ordered abelian group.

The *valuation ring* or *ring of integers of the valued field* (K, v) is the subring

$$\mathcal{O}_v := \{x \in K \mid v(x) \geq 0\},$$

where 0 is the neutral element of Γ. This ring $\mathcal{O}_v$ is a local ring with maximal ideal

$$\mathfrak{m}_v := \{x \in K \mid v(x) > 0\}.$$

Its *residue field* is defined as $\kappa_v := \mathcal{O}_v/\mathfrak{m}_v$.

Definition A.2.2 (Integral element over a ring)**.** Let R be a commutative ring with unity.

1. Let A be an R-algebra. An element $a \in A$ is called *integral over R* if there exists a monic polynomial

$$p(x) = x^n + r_{n-1}x^{n-1} + \cdots + r_1 x + r_0$$

with coefficients $r_i \in R$ such that

$$p(a) = a^n + r_{n-1}a^{n-1} + \cdots + r_1 a + r_0 = 0 \quad \text{in } A.$$

2. Let A be an extension ring of R. The ring extension A/R is called an *integral extension* if every element of A is integral over R.

3. Let A be an extension ring of R. The *integral closure of R in A*, sometimes denoted by $\widetilde{A}$, is the subring
$$\widetilde{A} = \{a \in A : a \text{ is integral over } R\}.$$
We say that R is integrally closed in A if $\widetilde{A}$ coincides with A (considered as a subring of R).
4. Let R be an integral domain with field of fractions $K = \mathrm{Frac}(R)$. We say that R is *integrally closed* if it is integrally closed as a subring of K.

Definition A.2.3 (Function field)**.** Let k be a field (Definition A.1.3). A field K is called a *function field over k* (or a *function field in n variables over k*) if K is a finitely generated field extension of k with transcendence degree n over k, where $n \geq 1$. Equivalently, the function field in n variables over k is isomorphic to the fraction field of the polynomial ring $k[x_1, \ldots, x_n]$ where $x_1, \ldots, x_n$ are indeterminate variables. Accordingly, the function field in n variables over k is often denoted as $k(x_1, \ldots, x_n)$.

When $n = 1$, we say that K is a *function field of one variable over $k* or an *algebraic function field over k* and denote it by $k(x)$.

Definition A.2.4. Let $f : X \to Y$ be a morphism of schemes (Definition 1.2.1). For any geometric point (either algebraic or separable) $\bar{y} : \mathrm{Spec}(\Omega) \to Y$, the *geometric fiber of f at $\bar{y}$*, denoted $X_{\bar{y}}$, is the base change (Definition A.2.5):
$$X_{\bar{y}} := X \times_Y \mathrm{Spec}(\Omega)$$
Note that this depends on the choice of the geometric point $\bar{y}$, not just on the image point $y \in Y$.

Definition A.2.5. Let $\mathcal{C}$ be a category. Let $f : X \to Y$ be a morphism in $\mathcal{C}$, and let $g : Y' \to Y$ be any morphism in $\mathcal{C}$ such that the fiber product $X \times_Y Y'$ exists. The *base change of f by g* is the morphism $f' : X \times_Y Y' \to Y'$ obtained by the pullback of f along g. In particular,
$$\begin{array}{ccc} X \times_Y Y' & \xrightarrow{p_1} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$
is a Cartesian diagram (or pullback square) in $\mathcal{C}$.

Definition A.2.6. If X is a scheme over a field k, we say X is *geometrically connected over k* if the structure morphism $X \to \mathrm{Spec}(k)$ has geometrically connected fibers; equivalently, if $X \times_k \mathrm{Spec}(\bar{k})$ is connected.

A.3. **Sheaves.**

Definition A.3.1 (Sheaf on a site)**.** Let $(\mathcal{C}, J)$ be a site. Let $\mathcal{A}$ be a (large) category.

1. A presheaf $\mathcal{F} : \mathcal{C}^{\mathrm{op}} \to \mathcal{A}$ is called a *sheaf on the site $(\mathcal{C}, J)$ valued in $\mathcal{A}$* if, for every object U of $\mathcal{C}$ and every covering sieve $S \in J(U)$, the limit
$$\varprojlim_{(V \to U) \in (\mathcal{D}_S)^{\mathrm{op}}} \mathcal{F}|_{\mathcal{D}_S}(V),$$

exists and the canonical natural morphism

$$\mathcal{F}(U) \to \varprojlim_{(V \to U) \in (\mathcal{D}_S)^{\mathrm{op}}} \mathcal{F}|_{\mathcal{D}_S}(V)$$

is an isomorphism. Here, $\mathcal{D}_S \hookrightarrow \mathcal{C}/U$ is the full downward-closed subcategory such that $\mathrm{Ob}(\mathcal{D}_S) = \{(f : V \to U) : f \in S(V)\}$,

In particular, when we are working with a Grothendieck pretopology K on a category $\mathcal{C}$, we may speak of sheaves on the site whose Grothendieck topology is the one generated by K.

2. Given sheaves $\mathcal{F}, \mathcal{G} : \mathcal{C}^{\mathrm{op}} \to \mathcal{A}$ on the site $(\mathcal{C}, J)$, a *morphism between the sheaves* is a morphism between $\mathcal{F}$ and $\mathcal{G}$ as presheaves.
3. Let U be a universe. A *U-sheaf* typically refers to a U-presheaf that is a sheaf for a U-site. In other words, a U-sheaf is a sheaf on a site whose underlying category is U-locally small and which has a U-small topologically generating family such that the sheaf is valued in U-sets.
4. The *sheaf category/category of $\mathcal{A}$-valued sheaves on $\mathcal{C}$* is the (large) category defined as the full subcategory of $\mathrm{PreShv}(\mathcal{C}, \mathcal{A})$ whose objects are the sheaves on $\mathcal{C}$ with values in $\mathcal{A}$. Common notations for the sheaf category include $\mathrm{Shv}(\mathcal{C}, \mathcal{A})$, $\mathrm{Shv}(\mathcal{C}, J, \mathcal{A})$, $\mathrm{Sh}(\mathcal{C}, \mathcal{A})$, $\mathrm{Sh}(\mathcal{C}, J, \mathcal{A})$. If the value category $\mathcal{A}$ is clear from context, then notations such as $\mathrm{Shv}(\mathcal{C})$, $\mathrm{Shv}(\mathcal{C}, J)$, $\mathrm{Sh}(\mathcal{C})$, $\mathrm{Sh}(\mathcal{C}, J)$ are also common.

Definition A.3.2 (Sheaf on a topological space)**.** Let X be a topological space, let $\mathcal{D}$ be a category; a posteori, it is necessary for $\mathcal{D}$ to have a terminal object for the following definition to be non-void. Let $\mathcal{F}$ be a presheaf valued in $\mathcal{D}$ on X. Then $\mathcal{F}$ is a *sheaf* if it satisfies the following additional condition (known as the *sheaf axioms*):

For every open set $U \subseteq X$ and every open cover $\{U_i\}_{i \in I}$ of U, let $\mathcal{J}$ be the diagram in the category of opens of U consisting of the inclusions $U_i \cap U_j \hookrightarrow U_i$ for all $i, j \in I$. Then $\mathcal{F}$ is a sheaf if the limit of the diagram $\mathcal{F} \circ \mathcal{J}$ exists in $\mathcal{D}$ and the natural morphism

$$\mathcal{F}(U) \to \lim_{j \in \mathcal{J}} \mathcal{F}(j)$$

is an isomorphism. More precisely, $\mathcal{J} : J \to \mathrm{Open}(U)$ should be the functor whose index category J consists of

1. An object i for every $i \in I$ and an object (i, j) for every pair $i, j \in I$,
2. Morphisms $p_1 : (i, j) \to i$ and $p_2 : (i, j) \to j$ for every pair $i, j \in I$

and which sends the objects and morphisms as follows:

1. $\mathcal{J}(i) = U_i$
2. $\mathcal{J}(i, j) = U_i \cap U_j$
3. $\mathcal{J}(p_1) : U_i \cap U_j \hookrightarrow U_i$
4. $\mathcal{J}(p_2) : U_i \cap U_j \hookrightarrow U_j$.

In particular, taking $U = \emptyset$ and taking the empty open cover of the empty set, $\mathcal{F}(\emptyset)$ must be the terminal object of $\mathcal{D}$

In the case that $\mathcal{D}$ admits all small limits, the sheaf condition is equivalent to the following: For every open set $U \subset X$ and every open cover $\{U_i\}_{i\in I}$ of U, the following equalizer diagram is exact:

$$\mathcal{F}(U) \to \prod_{i\in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j).$$

Here, the morphism $\mathcal{F}(U) \to \prod_{i\in I} \mathcal{F}(U_i)$ and the two morphisms $\prod_{i\in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j)$ are induced by the restriction maps $\mathcal{F}(U) \to \mathcal{F}(U_i)$ and $\mathcal{F}(U_i) \to \mathcal{F}(U_i \cap U_j)$.

In the case that $\mathcal{D}$ is some subcategory of the category of sets, the sheaf condition is equivalent to the following: For every open set $U \subseteq X$ and every open cover $\{U_i\}_{i\in I}$ of U,

- (Locality) If $s, t \in \mathcal{F}(U)$ are such that $s|_{U_i} = t|_{U_i}$ for all i, then $s = t$.
- (Gluing) If for each i there is $s_i \in \mathcal{F}(U_i)$ such that for all i, j one has $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$, then there exists a unique $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i.

Equivalently, a sheaf on a topological space X may be defined as a sheaf on (Definition A.3.1) the site of opens on X.

Definition A.3.3. 1. Let $\mathcal{C}$ be a site, and let $\mathcal{A}$ and $\mathcal{B}$ be sheaves (Definition A.3.1) of (not necessarily commutative) rings on $\mathcal{C}$.

(a) An *$(\mathcal{A}, \mathcal{B})$-bimodule* (or a *bimodule over $(\mathcal{A}, \mathcal{B})$*) is a sheaf (Definition A.3.1) $\mathcal{M}$ of abelian groups on $\mathcal{C}$ equipped with a left $\mathcal{A}$-module structure given by a morphism of sheaves (Definition A.3.1) of sets

$$\lambda : \mathcal{A} \times \mathcal{M} \longrightarrow \mathcal{M},$$

and a right $\mathcal{B}$-module structure given by a morphism of sheaves of sets

$$\rho : \mathcal{M} \times \mathcal{B} \longrightarrow \mathcal{M},$$

such that the actions are compatible. Specifically, for every object U in $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the equality

$$\lambda_U(a, \rho_U(m, b)) = \rho_U(\lambda_U(a, m), b)$$

holds in $\mathcal{M}(U)$. In standard multiplicative notation where $\lambda(a, m)$ is denoted $a \cdot m$ and $\rho(m, b)$ is denoted $m \cdot b$, this condition is the associativity axiom

$$(a \cdot m) \cdot b = a \cdot (m \cdot b).$$

In particular, for every object $U \in \mathcal{C}$, the abelian group $\mathcal{M}(U)$ has the structure of an $\mathcal{A}(U) - \mathcal{B}(U)$-bimodule.

(b) Assume that a sheafification functor

$$\mathrm{PreShv}(\mathcal{C}, \mathbf{Rings}) \to \mathrm{Shv}(\mathcal{C}, \mathbf{Rings})$$

exists. For rings A and B, a *A-B-bimodule* or *bimodule over (A, B) on $\mathcal{C}$* is a bimodule over $(\underline{A}, \underline{B})$ where $\underline{A}$ and $\underline{B}$ are the constant sheaves on $\mathcal{C}$ with values A and B respectively.

(c) Let $\mathcal{M}$ and $\mathcal{N}$ be $(\mathcal{A}, \mathcal{B})$-bimodules. A *homomorphism of $(\mathcal{A}, \mathcal{B})$-bimodules* (or an *$(\mathcal{A}, \mathcal{B})$-linear morphism*) is a morphism of sheaves of abelian groups $f : \mathcal{M} \to \mathcal{N}$

such that for every object U of $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the following compatibility conditions hold:

$$f_U(a \cdot m) = a \cdot f_U(m) \quad \text{and} \quad f_U(m \cdot b) = f_U(m) \cdot b.$$

We denote the category of $(\mathcal{A}, \mathcal{B})$-bimodules, with morphisms being morphisms of sheaves of abelian groups compatible with both the left $\mathcal{A}$-action and the right $\mathcal{B}$-action, by $\mathcal{A}$-$\mathcal{B}$-Mod or sometimes by ${}_{\mathcal{A}}\mathsf{Mod}_{\mathcal{B}}$ (♠ TODO: talk about how bimodules can be identifies with left/right modules)

2. Let $(\mathcal{C}, J)$ be a site. Let $\mathcal{O}$ be a sheaf of (not necessarily commutative) rings on $(\mathcal{C}, J)$ (Definition A.3.1), i.e. $((\mathcal{C}, J), \mathcal{O})$ is a ringed site.
   (a) An *(left/right/two-sided) $\mathcal{O}$-module* consists of the following data:
   - A sheaf $\mathcal{F}$ of abelian groups on $(\mathcal{C}, J)$,
   - for every object $U \in \mathcal{C}$, the structure of an (left/right/two-sided) $\mathcal{O}(U)$-module on $\mathcal{F}(U)$,

   such that for every morphism $f : V \to U$ in $\mathcal{C}$, the restriction map

   $$\rho_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$$

   is $\mathcal{O}(U)$-linear when the $\mathcal{O}(U)$-action on $\mathcal{F}(V)$ is defined via the natural ring homomorphism

   $$\mathcal{O}(U) \to \mathcal{O}(V)$$

   induced by f.

   (b) Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}$-modules (Definition A.3.3).
   A *morphism of $\mathcal{O}$-modules* $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves (Definition A.3.1) of abelian groups such that, for every object $U \in \mathcal{C}$, the component map

   $$\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$$

   is $\mathcal{O}(U)$-linear, i.e. it satisfies

   $$\varphi_U(r \cdot s) = r \cdot \varphi_U(s) \quad \text{for all } r \in \mathcal{O}(U),\ s \in \mathcal{F}(U).$$

   The collection of all $\mathcal{O}$-modules together with their morphisms of $\mathcal{O}$-modules forms the *category of $\mathcal{O}$-modules*, denoted $\mathbf{Mod}(\mathcal{O})$. The Hom functors $\mathrm{Hom}_{\mathbf{Mod}(\mathcal{O})}$ of this category may be denoted by $\mathrm{Hom}_{\mathcal{O}}$.

   In case that a sheafification functor

   $$\mathrm{PreShv}(\mathcal{C}, \mathbf{Rings}) \to \mathrm{Shv}(\mathcal{C}, \mathbf{Rings})$$

   exists, a left, right, two-sided $\mathcal{O}$-module (and morphisms thereof) is equivalent to a $(\mathcal{O}, \mathbb{Z})$-bimodule, $(\mathbb{Z}, \mathcal{O})$-bimodule, and $(\mathcal{O}, \mathcal{O})$-bimodule (and morphisms thereof) respectively, where $\mathbb{Z}$ is the constant sheaf of the integer ring $\mathbb{Z}$. In particular, the category of $(\mathbb{Z}, \mathbb{Z})$-bimodules is equivalent to the category of sheaves of abelian groups on $\mathcal{C}$.

Definition A.3.4. Let X be a topological space. Let $\mathcal{A}$ and $\mathcal{B}$ be sheaves (Definition A.3.2) of (not necessarily commutative) rings on X.

1. An *$(\mathcal{A}, \mathcal{B})$-bimodule* is a sheaf $\mathcal{M}$ of abelian groups on X equipped with a left $\mathcal{A}$-module structure given by a morphism of sheaves of sets

$$\lambda : \mathcal{A} \times \mathcal{M} \longrightarrow \mathcal{M},$$

and a right $\mathcal{B}$-module structure given by a morphism of sheaves of sets

$$\rho : \mathcal{M} \times \mathcal{B} \longrightarrow \mathcal{M},$$

such that the actions are compatible. Specifically, for every open subset $U \subseteq X$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the equality

$$(a \cdot m) \cdot b = a \cdot (m \cdot b)$$

holds in $\mathcal{M}(U)$.

In particular, for every open subset $U \subseteq X$, the abelian group $\mathcal{M}(U)$ has the structure of an $\mathcal{A}(U) - \mathcal{B}(U)$-bimodule.

2. Let $\mathcal{M}$ and $\mathcal{N}$ be $(\mathcal{A}, \mathcal{B})$-bimodules. A *homomorphism of $(\mathcal{A}, \mathcal{B})$-bimodules* is a morphism of sheaves of abelian groups $f : \mathcal{M} \to \mathcal{N}$ such that for every open subset $U \subseteq X$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the following compatibility conditions hold:

$$f_U(a \cdot m) = a \cdot f_U(m) \quad \text{and} \quad f_U(m \cdot b) = f_U(m) \cdot b.$$

   We denote the category of $(\mathcal{A}, \mathcal{B})$-bimodules by $\mathcal{A}$-$\mathcal{B}$-Mod.

3. Let $\mathcal{O}$ be a sheaf of (not necessarily commutative) rings on X, i.e., $(X, \mathcal{O})$ is a ringed space.
   (a) An *(left/right/two-sided) $\mathcal{O}$-module* consists of the following data:
      - A sheaf $\mathcal{F}$ of abelian groups on X,
      - for every open subset $U \subseteq X$, the structure of an (left/right/two-sided) $\mathcal{O}(U)$-module on $\mathcal{F}(U)$,

      such that for every inclusion of open subsets $V \subseteq U$, the restriction map

$$\rho_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$$

      is $\mathcal{O}(U)$-linear when the $\mathcal{O}(U)$-action on $\mathcal{F}(V)$ is defined via the natural ring homomorphism

$$\mathcal{O}(U) \to \mathcal{O}(V)$$

      induced by the inclusion.
   (b) Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}$-modules. A *morphism of $\mathcal{O}$-modules* is a morphism of sheaves of abelian groups $\varphi : \mathcal{F} \to \mathcal{G}$ such that, for every open subset $U \subseteq X$, the component map

$$\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$$

      is $\mathcal{O}(U)$-linear.
      The collection of all $\mathcal{O}$-modules together with their morphisms forms the *category of $\mathcal{O}$-modules*, denoted by $\mathbf{Mod}(\mathcal{O})$.

These notions are special cases of those of Definition A.3.3, where the site is taken to be the category of open subsets of X with the usual Grothendieck topology.

Definition A.3.5. Let $((\mathcal{C}, J), \mathcal{O})$ be a ringed site, where $\mathcal{O}$ is a sheaf (Definition A.3.1) of rings on the site $(\mathcal{C}, J)$. (♠ TODO: This needs some genuine fixing to be definable on a general site, not just one given by a pretopology)

1. Let I be an indexing set. The *free sheaf of $\mathcal{O}$-modules of rank I* (or simply a *free sheaf*), denoted by $\mathcal{O}^{\oplus I}$, is the sheaf associated to the presheaf $U \mapsto \mathcal{O}(U)^{\oplus I}$. If I is finite with cardinality n, we usually write $\mathcal{O}^{\oplus n}$.
2. Let $\mathcal{A}$ be a (large) category, and let A be an object of $\mathcal{A}$. Assume that a sheafification functor (♠ TODO: If such a sheafification functor exist, does a sheafification functor exist when restricted to an object U?)

$$a : \mathrm{PreShv}(\mathcal{C}, J, \mathcal{A}) \to \mathrm{Shv}(\mathcal{C}, J, \mathcal{A})$$

(Definition A.3.1) exists. Let U be an object of $\mathcal{C}$.

An $\mathcal{O}$-module (Definition A.3.3) $\mathcal{F}$ is called *locally free of rank n on U* (for an integer $n \geq 0$) if there exists a covering sieve $\{U_i \to U\}_{i \in I}$ such that for each i, the restriction $\mathcal{F}|_{U_i}$ is isomorphic to the free sheaf $(\mathcal{O}|_{U_i})^{\oplus n}$ as an $\mathcal{O}|_{U_i}$-module.

We might call a locally free $\mathcal{O}$-module of rank n an *algebraic vector bundle of rank n*. A *(algebraic) line bundle* or *invertible sheaf* or is then an algebraic vector bundle of rank 1.

Definition A.3.6 (Tensor product of sheaves of bimodules)**.** Let $\mathcal{C}$ be a site. Let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be (not necessarily commutative) sheaves (Definition A.3.1) of rings on $\mathcal{C}$. Let $\mathcal{M}$ be an $(\mathcal{R}, \mathcal{S})$-bimodule (Definition A.3.3), and let $\mathcal{N}$ be an $(\mathcal{S}, \mathcal{T})$-bimodule.

(♠ TODO: Consolidate with)

1. Define the *presheaf tensor product* $\mathcal{M} \otimes_{p,\mathcal{S}} \mathcal{N}$ to be the presheaf of abelian groups given by

$$U \longmapsto \mathcal{M}(U) \otimes_{\mathcal{S}(U)} \mathcal{N}(U).$$

As a presheaf, it has the structure of a $\mathcal{R} - \mathcal{T}$-bimodule (♠ TODO: bimodule for a presheaf).
2. Assume that a sheafification functor

$$\mathrm{Shv}(\mathcal{C}, \mathbf{Sets}) \to \mathrm{PreShv}(\mathcal{C}, \mathbf{Sets})$$

exists. The *tensor product of sheaves $\mathcal{M}$ and $\mathcal{N}$*, denoted $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$, is defined to be the sheafification of the presheaf $\mathcal{M} \otimes_{p,\mathcal{S}} \mathcal{N}$. This tensor product becomes naturally an $(\mathcal{R}, \mathcal{T})$-bimodule.

The left $\mathcal{R}$-action and right $\mathcal{T}$-action are induced by the presheaf actions defined section-wise by

$$r \cdot (m \otimes n) = (r \cdot m) \otimes n,$$
$$(m \otimes n) \cdot t = m \otimes (n \cdot t),$$

for all $r \in \mathcal{R}(U)$, $t \in \mathcal{T}(U)$, $m \in \mathcal{M}(U)$, and $n \in \mathcal{N}(U)$. The sheafification functor preserves these actions, yielding the required bimodule structure on $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$.

Inductively, given sheaves of rings $\mathcal{R}_0, \ldots, \mathcal{R}_k$ and $(\mathcal{R}_{i-1}, \mathcal{R}_i)$-bimodules $\mathcal{M}_i$ for $i = 1, \ldots, k$, we may speak of the tensor product

$$\mathcal{M}_0 \otimes_{\mathcal{R}_1} \mathcal{M}_1 \otimes_{\mathcal{R}_2} \cdots \otimes_{\mathcal{R}_{k-1}} \mathcal{M}_k.$$

Tensor products of sheaves are associative up to canonical isomorphism(♠ TODO:), allowing us to omit parentheses. This iterated tensor product carries a natural $(\mathcal{R}_0, \mathcal{R}_k)$-bimodule structure.

Given a sheaf of rings $\mathcal{R}$ and a two-sided $\mathcal{R}$-module $\mathcal{M}$, we may also speak of the *n-fold tensor product* $\mathcal{M}^{\otimes n} = \mathcal{M}^{\otimes_{\mathcal{R}} n}$.

Definition A.3.7 (Sheafified Hom)**.** Let $(\mathcal{D}, K)$ be a site that admits a small topologically generating family (e.g. by being essentially small) and let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be sheaves of rings on $(\mathcal{D}, K)$. Let $\mathcal{M}$ be a sheaf (Definition A.3.1) of $\mathcal{R}$-$\mathcal{S}$ bimodules (Definition A.3.3) and $\mathcal{N}$ be a sheaf of $\mathcal{R}$-$\mathcal{T}$ bimodules.

The *sheafified Hom*, denoted by notations such as $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, $\mathscr{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, or $\underline{\mathrm{Hom}}_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, is the sheaf of $\mathcal{S}$-$\mathcal{T}$ bimodules defined section-wise by:

$$\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})(U) = \mathrm{Hom}_{\mathrm{Sh}(\mathcal{D}/U, K|_U; \mathrm{Mod}(\mathcal{R}|_U))}(\mathcal{M}|_U, \mathcal{N}|_U).$$

(**??**) (**??**) (**??**)

Equivalently, it is the unique sheaf which satisfies the right adjoint property relative to the sheafified tensor product (**??**).

Definition A.3.8. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site. The *multiplicative group sheaf of the ringed site* is the sheaf (Definition A.3.1) $\mathcal{O}^\times$ of groups on $(\mathcal{C}, J)$ given by

$$\mathcal{O}^\times(U) = \mathcal{O}(U)^\times$$

whose group structure is the multiplication inherited from the ring $\mathcal{O}(U)$.

The sheaf is also called the *unit sheaf of the ringed site* and is also denoted by $\mathbb{G}_m$.

A.4. Galois modules and representations.

Definition A.4.1. Let G be a group and R a not necessarily commutative ring.

1. A *trivial G-module over R* or a *trivial $R[G]$-module* is an $R[G]$-module (say left) A on which G acts trivial, i.e. $ga = a$ for all $g \in G$ and $a \in A$.
2. Given a (left/right) R-module A, its *trivial G-module (over R)* is the $R[G]$-module structure whose underlying R-module is A and whose G-action is trivial. This construction yields additive functors

$$_R\mathrm{Mod} \to_{R[G]} \mathrm{Mod}$$

$$\mathrm{Mod}_R \to \mathrm{Mod}_{R[G]}$$

Definition A.4.2. Let R and S be topological rings. A *topological (R, S)-bimodule* is an (R, S)-bimodule M equipped with a topology such that:

1. M is a topological abelian group under addition.
2. The left action map $R \times M \to M$, defined by $(r, m) \mapsto rm$, is continuous.
3. The right action map $M \times S \to M$, defined by $(m, s) \mapsto ms$, is continuous.

Here, the product sets $R \times M$ and $M \times S$ are endowed with the product topology.

One also calls a topological module a *continuous module*.

A *left topological R-module* is a topological $(R, \mathbb{Z})$-bimodule, where $\mathbb{Z}$ carries the discrete topology. Similarly, a *right topological R-module* is a topological $(\mathbb{Z}, R)$-bimodule, where $\mathbb{Z}$ carries the discrete topology. Furthemore, a *two-sided topological R-module* is a topological (R, R)-bimodule.

The discrete topology on $\mathbb{Z}$ ensures that the required continuity of the $\mathbb{Z}$-action is satisfied by any topological abelian group M, as the map $\mathbb{Z} \times M \to M$ is continuous if and only if for each $n \in \mathbb{Z}$, the map $m \mapsto nm$ is continuous.

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